

Differentiable Electrochemical Simulation for Efficient Bayesian Inference



Jack Montgomery
St Anne's College
University of Oxford

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Abstract

This is a project on the application of gradient computation to the inverse problem in electrochemistry.

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Chapter 1

Introduction

Chapter 2

Electrochemical Systems

A typical electrochemical experiment uses an electrochemical cell to drive reduction and oxidation (redox) reactions at the surface of a working electrode. The cell consists of at least two electrodes immersed in an electrolyte solution: the *working electrode*, at whose surface the reaction of interest takes place, and a *counter electrode*, which completes the circuit. A third *reference electrode* is commonly included to provide a stable, known potential against which the potential of the working electrode is measured and controlled. A device called a *potentiostat* maintains the desired potential difference between the working and reference electrodes, drives the current through the circuit formed by the working and counter electrodes, and simultaneously measures that current.

The reaction occurring at the working electrode surface depends on the applied potential and the chemical species present in solution. In the simplest case, a solution-phase species undergoes a single-electron redox reaction:



where species A is reduced — that is, it gains one electron — to form species B. The reverse process, in which B loses an electron to regenerate A, is the corresponding oxidation. Throughout this chapter we restrict attention to this single-electron ($n = 1$) process and adopt the convention that a *positive* current corresponds to net reduction (cathodic current).

2.1 Voltammetry

Voltammetry is the class of electroanalytical techniques in which the potential applied to the working electrode is varied in a prescribed manner and the resulting current is recorded. Because the current is directly related to the rate of the electrochemical

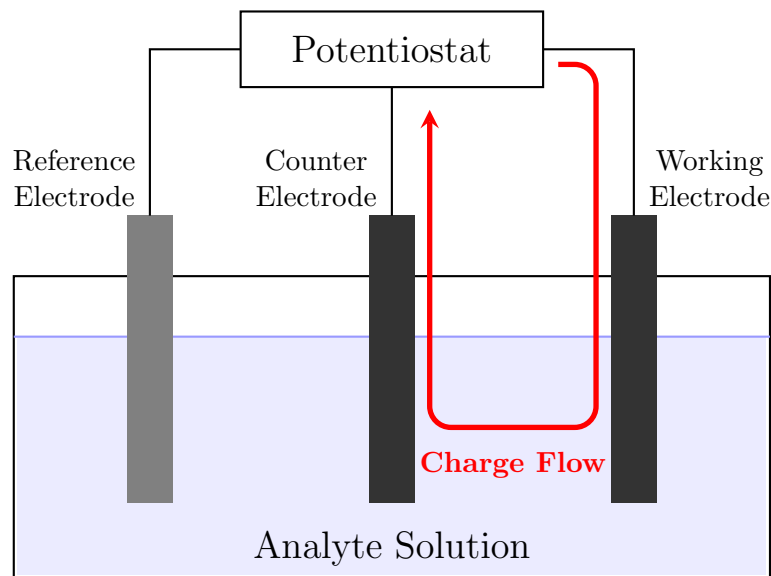


Figure 2.1: Schematic of a three-electrode electrochemical cell. The potentiostat controls the potential difference between the working and reference electrodes whilst driving current (red) through the circuit formed by the working and counter electrodes via the analyte solution.

reaction at the electrode surface, the current–potential curve — called a *voltammogram* — encodes information about the kinetics of the redox process as well as the transport of chemical species in solution.

In a cyclic voltammetry (CV) experiment the applied potential is swept linearly with time from a starting potential E_i , chosen so that the overpotential is insufficient to drive any significant reaction, towards more negative values (for a reduction) until a vertex potential E_v is reached. The vertex potential is chosen sufficiently beyond the formal potential of the redox couple that the reaction is driven to completion at the electrode surface; it serves only as the turning point of the sweep. The potential is then swept back to E_i , reversing the direction of electron transfer and reforming A. This triangular waveform is illustrated in the left panel of Fig. 2.2.

Throughout this process the current I is recorded. Plotting the current against the applied potential gives the voltammogram shown in the right panel of Fig. 2.2. The forward sweep produces a reduction (cathodic) peak as the potential moves through the range at which A is converted to B; the return sweep produces an oxidation (anodic) peak as B is reconverted to A.

The peak shape arises from competition between two rate-limiting processes. During the early part of the forward sweep the applied potential is positive relative to the formal potential of the redox couple, so there is insufficient driving force to reduce A

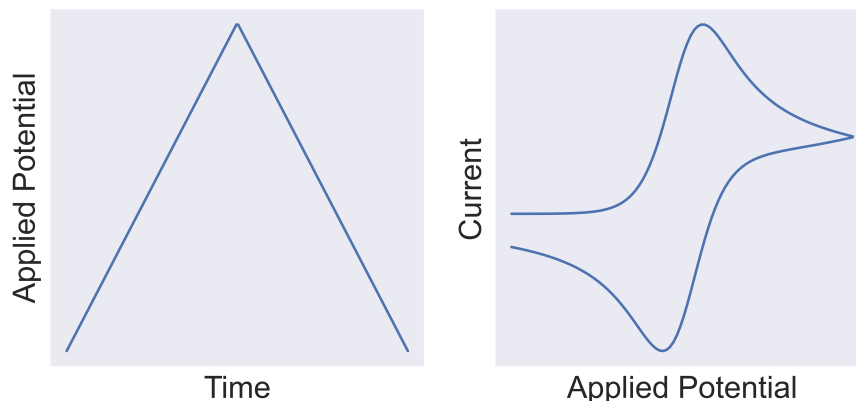


Figure 2.2: Left: the triangular potential waveform applied in a cyclic voltammetry experiment. Right: the resulting reversible reaction’s voltammogram showing the cathodic (reduction) peak which occurs on the forward sweep and the anodic (oxidation) peak which occurs on the reverse sweep.

and negligible current flows. As the potential is swept to more negative values the reduction reaction becomes progressively more thermodynamically favourable and the current increases: during this rising portion the system is under *kinetic control*, meaning that the rate of electron transfer at the electrode surface is the rate-limiting step. Eventually the concentration of A in the immediate vicinity of the electrode is depleted to near zero and the current reaches a peak. Beyond this point the current decays despite the continued increase in thermodynamic driving force, because the reaction has passed into *diffusional control*: the rate at which fresh A can be transported to the electrode by diffusion is now the limiting factor.

Cyclic Voltammetry is particularly useful because the position, shape, and relative heights of the peaks provide a signature of the electrochemical process. The potential at which each peak occurs is related to the formal potential E_f° , and the separation between the cathodic and anodic peaks carries information about the electrode kinetics, as discussed in Section 2.2.2.

The *scan rate* ν (V s^{-1}) is the constant rate at which the potential is swept, defined as

$$\nu = \frac{dE}{dt}. \quad (2.2)$$

The scan rate is an important experimental parameter: faster scans probe shorter timescales and can resolve kinetic processes that would otherwise reach equilibrium on a slower scan, whilst slower scans allow more time for diffusion and can reveal

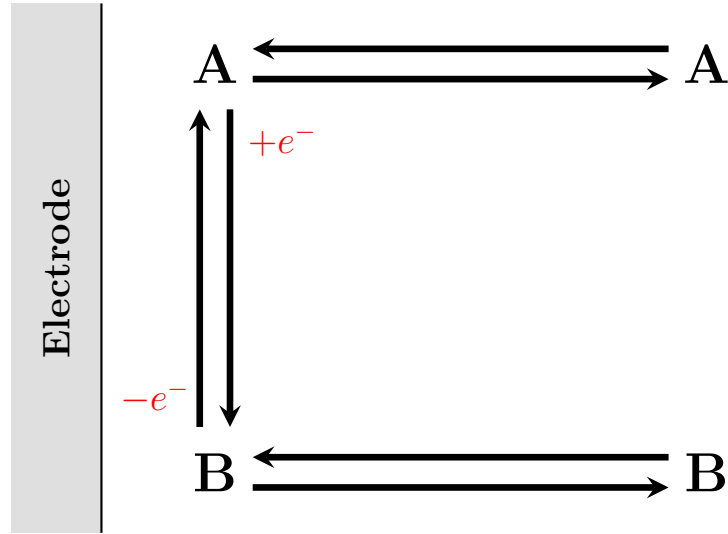


Figure 2.3: Schematic of the one-dimensional electrode reaction. Species A diffuses from the bulk solution towards the electrode surface, where it is reduced ($+e^-$) to form species B. The reverse oxidation ($-e^-$) converts B back to A. Species B diffuses away from the electrode into the bulk.

coupled chemical reactions.

2.2 Mathematical Description

We now formulate the governing equations for the system. The first assumption we make is that we have a “macro”-electrode: specifically, we consider a planar disc electrode of radius ϵ , whose characteristic dimension is much larger than the distance over which species can diffuse on the timescale of the experiment. As a consequence, edge effects are negligible and we can model only the diffusion perpendicular to the electrode surface, reducing the problem to one spatial dimension.

2.2.1 Diffusion: Fick’s Second Law

We model the transport of species in quiescent (unstirred) solution using Fick’s second law [2], which describes the evolution of a concentration field $c(x, t)$ due to diffusion. For the couple A/B on the planar electrode we have the system

$$\begin{aligned}\frac{\partial c_A}{\partial t} &= D_A \frac{\partial^2 c_A}{\partial x^2}, \\ \frac{\partial c_B}{\partial t} &= D_B \frac{\partial^2 c_B}{\partial x^2},\end{aligned}\tag{2.3}$$

where D_A and D_B ($\text{m}^2 \text{s}^{-1}$) are the diffusion coefficients of the respective species and x denotes the distance from the electrode surface. Each equation is a linear parabolic partial differential equation defined on the semi-infinite domain $x \geq 0$.

2.2.2 Fick's First Law and Electrode Kinetics

Fick's first law [2] states that the diffusive flux through a plane normal to the x -axis at a point x^* is proportional to the concentration gradient at that point:

$$j_{x^*} = -D \left(\frac{\partial c}{\partial x} \right) \Big|_{x=x^*}, \quad (2.4)$$

where the negative sign indicates that the flux is directed down the concentration gradient.

The rate at which the electrode reaction Eq. (2.1) proceeds is governed by the applied potential. The standard model for electrode kinetics is the Butler–Volmer equation [3], which relates the applied potential $E(t)$ to the net flux of A at the electrode surface:

$$j_{0,A} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} \right], \quad (2.5)$$

where k_0 (ms^{-1}) is the standard rate constant, $\alpha \in (0, 1)$ is the transfer coefficient, F is Faraday's constant, $\mathcal{R}$ is the universal gas constant, and $\mathcal{T}$ is the absolute temperature. The quantity $E(t) - E_f^0$ is the overpotential, measuring the deviation of the applied potential from the formal potential. The first exponential term represents the reduction contribution: a negative overpotential ($E(t) < E_f^0$) enhances this term and drives the conversion of A to B. The second exponential represents the oxidation contribution, which dominates when the overpotential is positive. In the limit of large k_0 the exponential terms force the surface concentrations towards thermodynamic equilibrium and the system exhibits reversible behaviour; when k_0 is small the reaction is kinetically limited and the flux is directly proportional to k_0 , giving irreversible behaviour.

Combining Eqs. (2.4) and (2.5) at the electrode surface ($x = 0$) yields a Neumann boundary condition for the diffusion equations:

$$-D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{\mathcal{R}\mathcal{T}} (E(t) - E_f^0)} \right]. \quad (2.6)$$

Conservation of mass at the electrode surface requires that whatever is consumed of A is produced as B, and vice versa. Since we consider a single-electron process with no homogeneous reactions in solution, this is simply stoichiometric conservation:

$$D_B \left(\frac{\partial c_B}{\partial x} \right) \Big|_{x=0} = -D_A \left(\frac{\partial c_A}{\partial x} \right) \Big|_{x=0}. \quad (2.7)$$

2.2.3 Boundary and Initial Conditions

At the start of the experiment we assume that A is at some bulk concentration c_A^* and that B is at bulk concentration c_B^* :

$$\begin{aligned} c_A(0, x) &= c_A^*, \\ c_B(0, x) &= c_B^*. \end{aligned} \tag{2.8}$$

Far from the electrode surface, the redox reaction has no effect on the concentrations, so the bulk values are recovered:

$$\begin{aligned} c_A(t, x) &\rightarrow c_A^* \quad \text{as } x \rightarrow \infty, \\ c_B(t, x) &\rightarrow c_B^* \quad \text{as } x \rightarrow \infty. \end{aligned} \tag{2.9}$$

2.2.4 Current

The primary experimental observable of a voltammetry experiment is the current measured by the potentiostat. At any instant the current for the single-electron reduction of species A is

$$I = F A_{\text{el}} j_{0,A}, \tag{2.10}$$

where $A_{\text{el}} = \pi\epsilon^2$ is the area of the disc electrode and $j_{0,A}$ is the net flux of species A at the electrode surface, which is proportional to the concentration gradient at that point by Fick's first law Eq. (2.4). Under our sign convention a positive $j_{0,A}$ corresponds to net consumption of A at the surface (reduction) and hence a positive current. Computing the current therefore reduces to evaluating how $j_{0,A}$ evolves in time.

2.3 Non-Dimensionalisation

The combination of Eqs. (2.3) and (2.6) to (2.9) constitutes an initial-boundary value problem. It is advantageous to solve a non-dimensional version of this problem: the results of a single non-dimensional simulation can be rescaled to any combination of physical parameters without rerunning the computation. We non-dimensionalise in the standard way, scaling concentrations by the bulk concentration of A, diffusion coefficients by D_A , lengths by the electrode radius ϵ , and potentials by the thermal voltage $\mathcal{RT}/F$.

We present a summary of the normalisation in Table 2.1.

Parameter	Normalisation
concentration	$C_j = c_j/c_A^*$
diffusion coefficient	$d_j = D_j/D_A$
spatial coordinate	$X = x/\epsilon$
time	$T = D_A t/\epsilon^2$
applied potential	$\theta = (F/\mathcal{RT}) E$
formal potential	$\theta_f = (F/\mathcal{RT}) E_f^0$
rate constant	$K_0 = k_0 \epsilon/D_A$
scan rate	$\sigma = (\epsilon^2/D_A)(F/\mathcal{RT}) \nu$
current	$J = I/(\pi \epsilon F D_A c_A^*)$

Table 2.1: Dimensionless normalisation of variables. Here ϵ is the radius of the disc electrode.

We define the dimensionless concentration, diffusion coefficient, spatial coordinate, and time as

$$C_j = \frac{c_j}{c_A^*}, \quad d_j = \frac{D_j}{D_A}, \quad X = \frac{x}{\epsilon}, \quad T = \frac{D_A t}{\epsilon^2}, \quad (2.11)$$

and the dimensionless rate constant as

$$K_0 = \frac{k_0 \epsilon}{D_A}. \quad (2.12)$$

We now substitute these into the governing equations, initial conditions, and boundary conditions.

2.3.1 Governing Equation

Considering the left-hand side of Eq. (2.3):

$$\frac{\partial c_j}{\partial t} = \frac{\partial c_j}{\partial C_j} \frac{\partial C_j}{\partial t} = c_A^* \frac{\partial C_j}{\partial t} = \frac{c_A^* D_A}{\epsilon^2} \frac{\partial C_j}{\partial T}.$$

Now the right-hand side:

$$D_j \frac{\partial^2 c_j}{\partial x^2} = D_j \frac{c_A^*}{\epsilon^2} \frac{\partial^2 C_j}{\partial X^2} = d_j \frac{c_A^* D_A}{\epsilon^2} \frac{\partial^2 C_j}{\partial X^2}.$$

Equating gives the dimensionless diffusion equation:

$$\frac{\partial C_j}{\partial T} = d_j \frac{\partial^2 C_j}{\partial X^2}. \quad (2.13)$$

2.3.2 Initial and Far-Field Conditions

The initial conditions become

$$\begin{aligned} C_A(0, X) &= 1, \\ C_B(0, X) &= C_B^*. \end{aligned} \quad (2.14)$$

In practice we cannot simulate a semi-infinite domain. Einstein's analysis of Brownian motion [4] shows that the root-mean-square displacement of a diffusing particle is $x_{\text{rms}} = \sqrt{2Dt}$. To ensure the outer boundary is sufficiently remote that it is unaffected by diffusion, we truncate the domain at $X_{\text{max}} = 6\sqrt{T_{\text{sim}}}$ in dimensionless form. The far-field boundary conditions are then

$$\begin{aligned} C_A(T, X_{\text{max}}) &= 1, \\ C_B(T, X_{\text{max}}) &= C_B^*. \end{aligned} \tag{2.15}$$

2.3.3 Electrode Surface Boundary

We define the dimensionless applied and formal potentials as

$$\begin{aligned} \theta(T) &= \frac{F}{\mathcal{RT}} E(t), \\ \theta_f &= \frac{F}{\mathcal{RT}} E_f^0. \end{aligned} \tag{2.16}$$

Substituting the dimensionless variables into Eqs. (2.6) and (2.7) and using the definition of K_0 from Eq. (2.12), the Butler–Volmer boundary condition and the mass conservation condition become

$$\begin{aligned} \left. \frac{\partial C_A}{\partial X} \right|_{X=0} &= K_0 [C_A(T, 0) e^{-\alpha(\theta(T)-\theta_f)} - C_B(T, 0) e^{(1-\alpha)(\theta(T)-\theta_f)}], \\ d_B \left. \frac{\partial C_B}{\partial X} \right|_{X=0} &= - \left. \frac{\partial C_A}{\partial X} \right|_{X=0}. \end{aligned} \tag{2.17}$$

2.3.4 Scan Rate

As defined in Eq. (2.2), the scan rate is the rate of change of potential with respect to time. We therefore define a dimensionless scan rate σ as the rate of change of dimensionless potential with respect to dimensionless time:

$$\sigma = \frac{d\theta}{dT}. \tag{2.18}$$

To relate σ to ν we note that

$$\nu = \frac{\partial E}{\partial t} = \frac{\mathcal{RT}}{F} \frac{\partial \theta}{\partial t} = \frac{\mathcal{RT} D_A}{F \epsilon^2} \frac{\partial \theta}{\partial T},$$

and hence

$$\sigma = \frac{F \epsilon^2}{\mathcal{RT} D_A} \nu. \tag{2.19}$$

2.3.5 Current

Since the model is now in dimensionless units we cannot compute the flux $j_{0,A}$ directly, but we can compute a dimensionless flux and relate the two. We define

$$J_{X^*,j} = -d_j \left(\frac{\partial C_j}{\partial X} \right)_{X^*}. \quad (2.20)$$

To recover the dimensional flux, we note that

$$j_j = -D_j \left(\frac{\partial c_j}{\partial x} \right)_{x^*} = -d_j \frac{c_A^* D_A}{\epsilon} \frac{\partial C_j}{\partial X} = \frac{c_A^* D_A}{\epsilon} J_j.$$

Substituting into Eq. (2.10) with $A_{\text{el}} = \pi \epsilon^2$ gives

$$I = \pi \epsilon F D_A c_A^* J_{0,A}. \quad (2.21)$$

When we run a simulation we record the dimensionless flux J as a function of the dimensionless potential θ . To obtain the physical voltammogram the results need only be scaled by the values of c_A^* , D_A , and ϵ . Consequently, the results of a single simulation apply to any set of these parameters; the simulation does not need to be rerun when, for example, the bulk concentration of A is changed.

2.4 Equal Diffusion Coefficients

For the remainder of this dissertation we assume that all diffusion coefficients are equal, i.e. $d_B = 1$. This assumption allows us to solve for the concentration of A alone and infer the concentration of B.

From the mass conservation condition Eq. (2.17), when $d_B = 1$ the surface flux condition simplifies to

$$\left. \frac{\partial C_B}{\partial X} \right|_{X=0} = - \left. \frac{\partial C_A}{\partial X} \right|_{X=0}. \quad (2.22)$$

Since the diffusion coefficients are equal and both species satisfy the same diffusion equation, the sum $C_A + C_B$ satisfies

$$\frac{\partial(C_A + C_B)}{\partial T} = \frac{\partial^2(C_A + C_B)}{\partial X^2} \quad (2.23)$$

with homogeneous Neumann conditions at $X = 0$ (by Eq. (2.22)) and constant far-field values. Together with the initial conditions, this implies that $C_A + C_B$ is constant everywhere in space and time. Any decrease in C_A is therefore matched by an equal increase in C_B , and vice versa, so that

$$C_A + C_B = 1 + C_B^*. \quad (2.24)$$

We henceforth drop the subscript and write $C \equiv C_A$, with $C_B = 1 + C_B^* - C$. The electrode boundary condition then reduces to

$$\left. \frac{\partial C}{\partial X} \right|_{X=0} = K_0 [C(T, 0) e^{-\alpha(\theta(T) - \theta_f)} - (1 + C_B^* - C(T, 0)) e^{(1-\alpha)(\theta(T) - \theta_f)}]. \quad (2.25)$$

For ease of notation we denote:

$$\begin{aligned} K_{\text{red}} &= K_0 \exp \left[-\alpha(\theta(T) - \theta_f) \right], \\ K_{\text{ox}} &= K_0 \exp \left[(1 - \alpha)(\theta(T) - \theta_f) \right]. \end{aligned} \quad (2.26)$$

These shorthand expressions are used extensively in the numerical discretisation that follows in next chapter.

Chapter 3

Numerical Methods

In this chapter we discretise the dimensionless initial-boundary value problem derived in the previous chapter and describe how the resulting linear system is solved at each timestep. The reader is assumed to be familiar with finite difference methods; we focus on the aspects specific to the electrochemical setting, namely the exponentially expanding spatial mesh and the incorporation of the Butler–Volmer boundary condition into the tridiagonal system.

For the remainder of this chapter we assume $C_B^* = 0$ (no product initially present in solution), so that the equal-diffusion-coefficient boundary condition Eq. (2.25) becomes

$$\left. \frac{\partial C}{\partial X} \right|_{X=0} = K_{\text{red}}(T) C(T, 0) - K_{\text{ox}}(T) (1 - C(T, 0)), \quad (3.1)$$

where K_{red} and K_{ox} were defined in Eq. (2.26) and depend on time through the applied potential $\theta(T)$.

3.1 Discretisation

3.1.1 Temporal Grid

We divide the simulation interval $[0, T_{\text{max}}]$ into m uniform steps of size

$$\Delta T = \frac{T_{\text{max}}}{m}, \quad (3.2)$$

so that the discrete time levels are $T^k = k \Delta T$ for $k = 0, 1, \dots, m$. Here T_{max} is determined by the scan rate and the potential window of the experiment: for a cyclic voltammogram sweeping from θ_i to θ_v and back, $T_{\text{max}} = \frac{2|\theta_v - \theta_i|}{\sigma}$.

3.1.2 Spatial Grid

The electrochemical reaction occurs at the electrode surface and its influence on the concentration profile is confined to a thin diffusion layer near $X = 0$. Far from the electrode the concentration remains at its bulk value and contributes little to the dynamics. It is therefore desirable to concentrate spatial resolution near the electrode surface whilst using coarser spacing in the far field. We achieve this with an exponentially expanding mesh controlled by two parameters: the initial spacing $h_0 > 0$ and an expansion factor $\omega > 1$.

We define n spatial nodes $\{X_0, X_1, \dots, X_{n-1}\}$ by

$$X_i = \begin{cases} \sum_{k=0}^{i-1} h_0 \omega^k, & 0 \leq i \leq n-2, \\ X_{\max}, & i = n-1, \end{cases} \quad (3.3)$$

with $X_0 = 0$ corresponding to the electrode surface. The local spacings about point i are

$$\Delta X_+ = X_{i+1} - X_i, \quad \Delta X_- = X_i - X_{i-1}. \quad (3.4)$$

For $0 \leq i \leq n-2$ the spacing grows geometrically as $\Delta X_+ = h_0 \omega^i$, so the mesh is finest at the electrode and coarsens exponentially into the bulk. The final point ($i = n-1$) is placed at $X_{\max} = 6\sqrt{T_{\max}}$ as discussed in the previous chapter, ensuring the far-field boundary does not influence the solution.

3.2 Finite Difference Scheme

We discretise the dimensionless diffusion equation Eq. (2.13) (with $d_j = 1$) using a backward Euler (fully implicit) scheme in time and a three-point central difference approximation for the second spatial derivative on the non-uniform mesh.

Writing $C_i^k \approx C(T^k, X_i)$, the scheme at an interior node $1 \leq i \leq n-2$ reads

$$\frac{C_i^k - C_i^{k-1}}{\Delta T} = \frac{1}{\frac{1}{2}(\Delta X_+ + \Delta X_-)} \left(\frac{C_{i+1}^k - C_i^k}{\Delta X_+} - \frac{C_i^k - C_{i-1}^k}{\Delta X_-} \right). \quad (3.5)$$

The implicit treatment evaluates the spatial operator at the new time level k , which makes the scheme unconditionally stable and avoids a restrictive CFL condition on the timestep.

3.2.1 Interior Coefficients

Rearranging Eq. (3.5) so that all unknowns at time level k appear on the left and the known value at level $k - 1$ appears on the right, we obtain

$$\alpha_i C_{i-1}^k + \beta_i C_i^k + \gamma_i C_{i+1}^k = C_i^{k-1}, \quad 1 \leq i \leq n - 2, \quad (3.6)$$

where the tridiagonal coefficients are

$$\begin{aligned} \alpha_i &= -\frac{2 \Delta T}{\Delta X_- (\Delta X_+ + \Delta X_-)}, \\ \gamma_i &= -\frac{2 \Delta T}{\Delta X_+ (\Delta X_+ + \Delta X_-)}, \\ \beta_i &= 1 - \alpha_i - \gamma_i. \end{aligned} \quad (3.7)$$

3.2.2 Electrode Boundary ($i = 0$)

At the electrode surface we approximate the spatial gradient in Eq. (3.1) with a two-point forward difference:

$$\frac{C_1^k - C_0^k}{h_0} = K_{\text{red}}^k C_0^k - K_{\text{ox}}^k (1 - C_0^k), \quad (3.8)$$

where K_{red}^k and K_{ox}^k denote the values of K_{red} and K_{ox} evaluated at the potential $\theta(T^k)$. Rearranging into the form $\beta_0 C_0^k + \gamma_0 C_1^k = b_0$ gives

$$\begin{aligned} \beta_0 &= 1 + h_0 K_{\text{red}}^k + h_0 K_{\text{ox}}^k, \\ \gamma_0 &= -1, \\ b_0 &= h_0 K_{\text{ox}}^k. \end{aligned} \quad (3.9)$$

3.2.3 Far-Field Boundary ($i = n - 1$)

The Dirichlet condition $C(T, X_{\text{max}}) = 1$ is imposed directly:

$$\begin{aligned} \beta_{n-1} &= 1, \\ b_{n-1} &= 1, \end{aligned} \quad (3.10)$$

with $\alpha_{n-1} = \gamma_{n-1} = 0$.

3.2.4 Solution of the Tridiagonal System

At each time level k the discretisation produces a tridiagonal linear system with lower diagonal $(\alpha_1, \dots, \alpha_{n-1})$, main diagonal $(\beta_0, \dots, \beta_{n-1})$, and upper diagonal $(\gamma_0, \dots, \gamma_{n-2})$.

Because the electrode boundary coefficients depend on the applied potential, the first row of the system is reassembled at each timestep; the interior coefficients are time-independent and can be precomputed.

This system is solved in $O(n)$ operations using the Thomas algorithm (tridiagonal LU factorisation), making the cost per timestep linear in the number of spatial nodes.

3.3 The Forward Solver as a Mapping

The implicit finite difference solver computes the concentration field at every node in the spatial-temporal mesh. However, the experimentally observable quantity is not the concentration profile but the current, which is proportional to the surface flux $J_{0,A}$ recorded at each timestep. It is therefore natural to regard the solver as a mapping from the set of model parameters to the predicted dimensionless current response.

We define the *forward map*

$$\mathcal{P}: \Phi \rightarrow \mathcal{J}, \quad \phi \mapsto \mathcal{P}(\phi), \quad (3.11)$$

where $\Phi \subseteq \mathbb{R}^3$ is the parameter space with $\phi = (\alpha, K_0, \theta_f)$, and $\mathcal{J}$ is the space of dimensionless flux trajectories $J_{0,A}(T)$ evaluated at the discrete time levels. Given a parameter vector ϕ , a single evaluation of $\mathcal{P}$ consists of assembling and solving the tridiagonal system at each of the m timesteps and extracting the surface flux at each step.

Although the electrode boundary condition is discretised with a two-point approximation (to preserve the tridiagonal structure), the surface flux used to compute the current is evaluated with a higher-order three-point formula. Writing $h_1 = X_1 - X_0$ and $h_2 = X_2 - X_0$, we fit a quadratic through (X_0, C_0^k) , (X_1, C_1^k) , (X_2, C_2^k) and differentiate at X_0 to obtain

$$J_{0,A}^k = - \left(\frac{\partial C}{\partial X} \right) \Big|_{X=0} \approx - \frac{h_2^2 (C_1^k - C_0^k) - h_1^2 (C_2^k - C_0^k)}{h_1 h_2 (h_2 - h_1)}. \quad (3.12)$$

This forward map is the central object in the inference problems discussed in the next chapter: given experimental data, we seek the parameters ϕ for which $\mathcal{P}(\phi)$ best reproduces the observed current response.

3.4 Effect of Butler–Volmer Parameters

In the electron-transfer reaction we have been discussing there are three parameters of interest: the transfer coefficient α , the dimensionless rate constant K_0 , and the

dimensionless formal potential θ_f . Each parameter has a distinct qualitative effect on the shape of the voltammogram. We illustrate these effects by varying each parameter in turn about a quasi-reversible baseline in Fig. 3.1.

Increasing the transfer coefficient α enhances the cathodic peak current whilst diminishing the anodic peak; decreasing α has the opposite effect. This asymmetry arises because α weights the reduction and oxidation exponentials in the Butler–Volmer equation: a larger α favours the cathodic branch at the expense of the anodic branch. The dimensionless rate constant K_0 has a symmetrising effect on the voltammogram. As K_0 increases, the cathodic and anodic peaks sharpen, grow in magnitude, and move closer together, reflecting the transition from kinetically limited to near-reversible behaviour. Finally, the formal potential θ_f translates the entire voltammogram along the potential axis without altering the peak shapes, since it simply shifts the point about which the overpotential is measured.

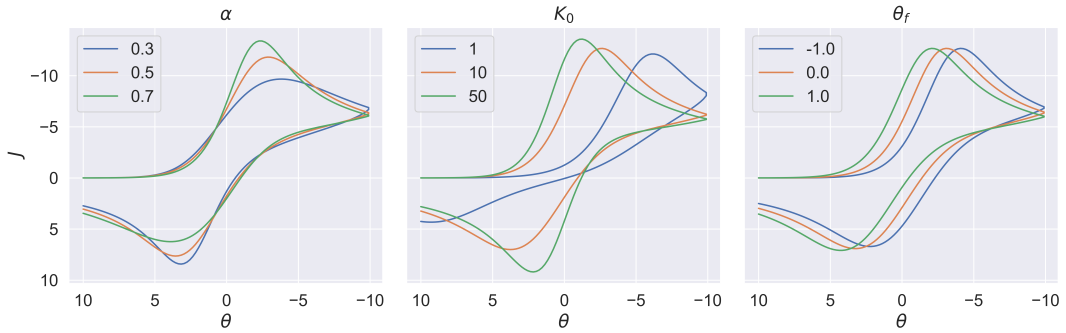


Figure 3.1: Effect of each Butler–Volmer parameter on the dimensionless voltammogram for a quasi-reversible reaction. In each panel one parameter is varied while the others are held fixed. Left: varying α changes the relative heights of the cathodic and anodic peaks. Centre: increasing K_0 symmetrises and sharpens the voltammogram. Right: varying θ_f translates the peaks along the potential axis.

The interplay between these parameters is important to note. Because changes in one parameter can produce features in the voltammogram that partially resemble changes in another, there is potential for compensating errors in parameter estimation. For instance, a rate constant that is too small — which broadens the peaks and shifts them apart — could be partially offset by an adjustment in the formal potential that translates the peaks back towards the observed positions. Such parameter correlations suggest that a point estimate obtained by optimisation alone may not capture the full uncertainty in the inferred values.

The situation is compounded in the reversible limit. Figure 3.2 shows the effect of varying α at $K_0 = 200$, well into the reversible regime. The three voltammograms for $\alpha = 0.3, 0.5, 0.7$ are nearly indistinguishable: when the rate constant is large the exponential terms in the Butler–Volmer equation force the surface concentrations to their thermodynamic equilibrium values regardless of α , and the transfer coefficient effectively drops out of the problem. In this regime, any attempt to estimate α from the voltammogram alone — whether by maximum likelihood or other point-estimation methods — will yield an essentially uninformative result.

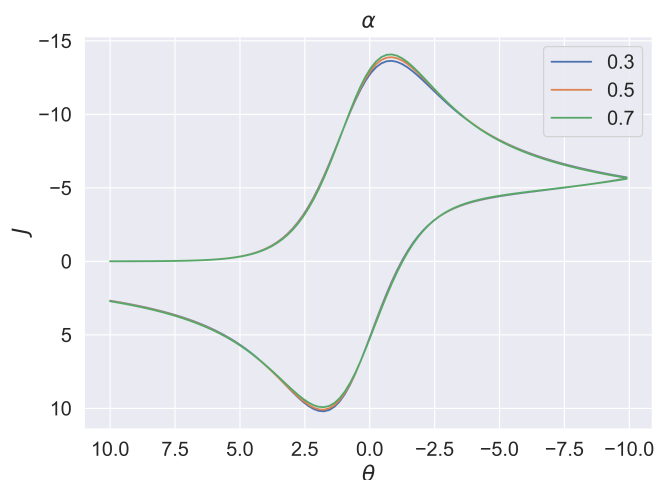


Figure 3.2: Effect of the transfer coefficient α on the voltammogram in the reversible regime ($K_0 = 200$). The three curves are nearly indistinguishable, demonstrating that α has negligible influence on the voltammogram when the reaction kinetics are fast.

The combination of parameter correlations in the quasi-reversible regime, the practical non-identifiability of α in the reversible regime, and the inherent noise present in experimental data means that estimating the Butler–Volmer parameters is a problem better suited to a probabilistic framework than to a purely optimisation-based approach. A probabilistic treatment naturally quantifies the uncertainty in each parameter and reveals correlations between them, providing a more complete picture than a single point estimate. We develop this approach in the following chapter.

Chapter 4

Bayesian Approach to the Inverse Problem

In general, data obtained from a voltammetry experiment will be noisy and subject to experimental error. Further, as discussed in Section 3.4, there may be multiple configurations of parameters that produce similar current waveforms, and certain parameters may have negligible influence on the voltammogram in particular kinetic regimes. This makes the problem of recovering the parameters of the experiment given the observed data — the *inverse problem* — *ill-posed*: a unique, stable solution may not exist. For this reason, we take a Bayesian approach and seek probability distributions over the parameters rather than point estimates. This framework naturally quantifies the uncertainty in each parameter and reveals correlations between them.

4.1 Noise Model

We model the observed dimensionless flux data as the output of the forward map $\mathcal{P}$ corrupted by additive Gaussian noise. Let $\mathbf{y} = (y_1, \dots, y_N)$ denote the N flux measurements recorded at discrete time levels. We assume

$$y_i = \mathcal{P}(\phi)_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \varsigma^2), \quad (4.1)$$

where $\mathcal{P}(\phi)_i$ is the i th component of the simulated flux trajectory for parameter vector $\phi = (\alpha, K_0, \theta_f)$, and ς^2 is the variance of the measurement noise, which we treat as a known constant throughout. We use ς rather than σ to avoid confusion with the dimensionless scan rate defined in the previous chapters.

An example of such noisy data is shown in Fig. 4.1.

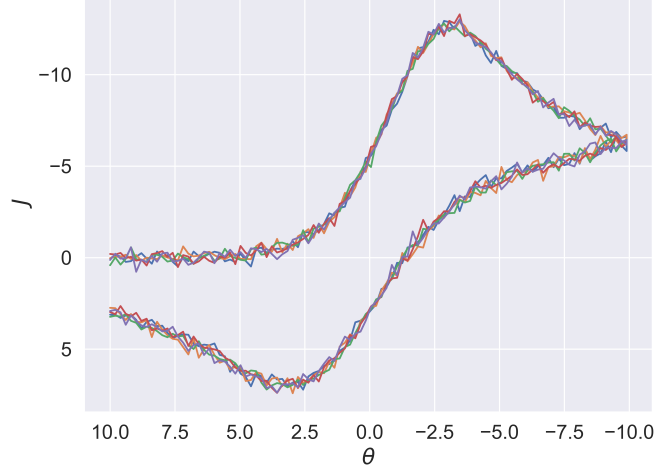


Figure 4.1: Synthetic voltammogram data generated by evaluating the forward map $\mathcal{P}$ with dimensionless scan rate $\sigma = 1000$ over the potential window $\theta_i = 10$ to $\theta_v = -10$, then adding i.i.d. Gaussian noise with $\varsigma = 0.25$. The noisy observations (points) are the data $\mathbf{y}$ from which the Butler–Volmer parameters are to be inferred.

4.2 Posterior Distribution

Bayes’ theorem relates the posterior distribution of the parameters given the data to the likelihood and prior:

$$p(\phi \mid \mathbf{y}) = \frac{L(\mathbf{y} \mid \phi) \pi(\phi)}{p(\mathbf{y})}, \quad (4.2)$$

where $L(\mathbf{y} \mid \phi)$ is the likelihood, $\pi(\phi)$ is the prior distribution, and $p(\mathbf{y}) = \int L(\mathbf{y} \mid \phi) \pi(\phi) d\phi$ is the marginal likelihood (or evidence), which serves as a normalising constant independent of ϕ .

4.2.1 Prior

We adopt an uninformative prior: a uniform distribution over physically reasonable bounds for each parameter,

$$\pi(\phi) \propto 1 \quad \text{for } \phi \in \Phi, \quad (4.3)$$

where $\Phi \subset \mathbb{R}^3$ is a bounded region encoding constraints such as $\alpha \in (0, 1)$ and $K_0 > 0$. Within Φ the prior is constant and therefore does not favour any particular parameter combination. Consequently, the posterior is proportional to the likelihood:

$$p(\phi \mid \mathbf{y}) \propto L(\mathbf{y} \mid \phi). \quad (4.4)$$

4.2.2 Likelihood

Under the noise model Eq. (4.1), the observations are conditionally independent given ϕ , and the likelihood takes the form

$$L(\mathbf{y} \mid \phi) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\varsigma^2}} \exp\left(-\frac{(y_i - \mathcal{P}(\phi)_i)^2}{2\varsigma^2}\right). \quad (4.5)$$

In practice we work with the log-likelihood. Taking the logarithm of Eq. (4.5) gives

$$\log L(\mathbf{y} \mid \phi) = -\frac{N}{2} \log(2\pi\varsigma^2) - \frac{1}{2\varsigma^2} \sum_{i=1}^N (y_i - \mathcal{P}(\phi)_i)^2. \quad (4.6)$$

Since ς is treated as known, the first term is a constant that does not depend on ϕ and can be dropped when comparing likelihoods at different parameter values. We therefore define the log-likelihood function used in all subsequent computations as

$$\ell(\phi) = -\frac{1}{2\varsigma^2} \sum_{i=1}^N (y_i - \mathcal{P}(\phi)_i)^2. \quad (4.7)$$

4.3 Markov Chain Monte Carlo

The posterior distribution Eq. (4.4) is known only up to a normalising constant, and the integral defining that constant is generally intractable for non-trivial forward maps. *Markov chain Monte Carlo* (MCMC) provides a numerical approach: rather than evaluating the posterior everywhere, we construct a Markov chain whose stationary distribution is the posterior and use the samples generated by the chain to approximate expectations and marginal distributions.

4.3.1 Markov Chain

An MCMC method produces a sequence of parameter vectors

$$\{\phi^{(0)}, \phi^{(1)}, \dots, \phi^{(T)}\} \subseteq \Phi, \quad (4.8)$$

where each $\phi^{(t+1)}$ depends only on the current state $\phi^{(t)}$ (the Markov property). Under regularity conditions — in particular, that the chain is irreducible and aperiodic — the ergodic theorem guarantees that the empirical distribution of the samples converges to the target posterior as $T \rightarrow \infty$.

4.3.2 Proposal Distribution

At each step the chain generates a *candidate* parameter vector from a *proposal distribution* $q(\cdot \mid \phi^{(t)})$:

$$\phi_{\text{cand}} \sim q(\cdot \mid \phi^{(t)}). \quad (4.9)$$

The proposal distribution is a design choice that determines how the chain explores the parameter space. Its form need not match the posterior, but the efficiency of the sampler depends heavily on how well the proposals reflect the local geometry of the target distribution.

4.3.3 Acceptance Criterion

Whether the chain moves to the candidate is determined by the *Metropolis–Hastings acceptance probability* [6]:

$$r = \min \left\{ \frac{p(\phi_{\text{cand}} \mid \mathbf{y}) q(\phi^{(t)} \mid \phi_{\text{cand}})}{p(\phi^{(t)} \mid \mathbf{y}) q(\phi_{\text{cand}} \mid \phi^{(t)})}, 1 \right\}. \quad (4.10)$$

A uniform random number $u \sim \text{Uniform}(0, 1)$ is drawn: if $u < r$ the candidate is accepted and $\phi^{(t+1)} = \phi_{\text{cand}}$; otherwise the chain remains at $\phi^{(t+1)} = \phi^{(t)}$. The ratio in Eq. (4.10) involves the posterior only through ratios, so the unknown normalising constant cancels – this is the key property that makes MCMC practical.

4.4 Random Walk Metropolis–Hastings

The *random walk Metropolis–Hastings* (RWMH) algorithm is a particular instance of the Metropolis–Hastings framework in which the proposal distribution is a symmetric Gaussian centred at the current state:

$$q(\phi_{\text{cand}} \mid \phi^{(t)}) = \mathcal{N}(\phi^{(t)}, \Sigma), \quad (4.11)$$

where $\Sigma \in \mathbb{R}^{3 \times 3}$ is a positive-definite covariance matrix that controls the size and orientation of the proposed steps. Since the Gaussian density is symmetric — $q(\phi_{\text{cand}} \mid \phi^{(t)}) = q(\phi^{(t)} \mid \phi_{\text{cand}})$ — the proposal ratio in Eq. (4.10) is unity, and the acceptance probability simplifies to

$$r = \min \left\{ \frac{p(\phi_{\text{cand}} \mid \mathbf{y})}{p(\phi^{(t)} \mid \mathbf{y})}, 1 \right\} = \min \{ \exp(\ell(\phi_{\text{cand}}) - \ell(\phi^{(t)})), 1 \}, \quad (4.12)$$

where the second equality uses Eq. (4.4) and the definition of the log-likelihood Eq. (4.7). In practice, we compute $\ell(\phi_{\text{cand}}) - \ell(\phi^{(t)})$ directly to avoid numerical overflow.

The choice of the proposal covariance Σ is critical to the efficiency of the sampler. If the steps are too small the chain explores the parameter space slowly and successive samples are highly correlated; if the steps are too large most proposals fall in regions of low posterior density and are rejected. A common heuristic is to tune Σ to achieve an acceptance rate of approximately 23% [5], which is asymptotically optimal for Gaussian targets in high dimensions.

4.4.1 Sampling Results

We apply the RWMH algorithm to synthetic data generated from the forward map with known true parameters and additive Gaussian noise with $\varsigma = 0.25$. Figure 4.2 shows the marginal posterior histograms for each parameter. All three marginals concentrate around the true parameter values, confirming that the sampler recovers the correct region of the parameter space. The marginal for K_0 and θ_f are sharply peaked, indicating that the data is informative for these parameters; the marginal for α is broader, consistent with the weaker sensitivity discussed in Section 3.4.

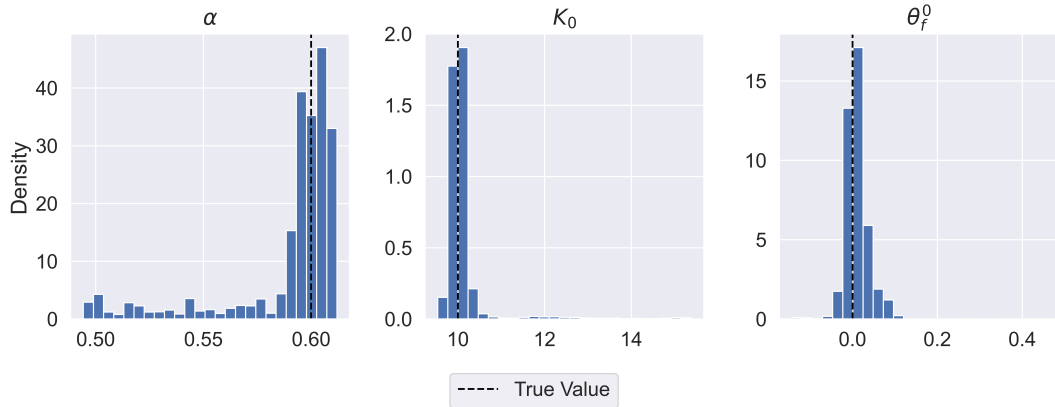


Figure 4.2: Marginal posterior histograms for the three Butler–Volmer parameters obtained by RWMH sampling on synthetic data with noise level $\varsigma = 0.25$. Dashed lines indicate the true parameter values used to generate the data. The posteriors for K_0 and θ_f are tightly concentrated around the true values, while the marginal for α is broader, reflecting its weaker influence on the voltammogram at this kinetic regime.

However, the histograms also exhibit extended tails, particularly for α . The origin of these tails is visible in Fig. 4.3, which shows the (α, K_0) projection of the chain coloured by sample index. The chain was initialised far from the high-density region and spends a substantial number of iterations traversing a low-probability ridge before settling near the true values. The standard practice in MCMC is to discard

these initial samples as *burn-in* and retain only the remainder as approximate draws from the posterior. In principle this is straightforward; in our setting, however, each likelihood evaluation requires a full run of the forward map $\mathcal{P}$, which involves solving the tridiagonal system at every timestep. The cost of generating each proposal is therefore non-trivial, and discarding a large burn-in period represents a significant waste of computational effort.

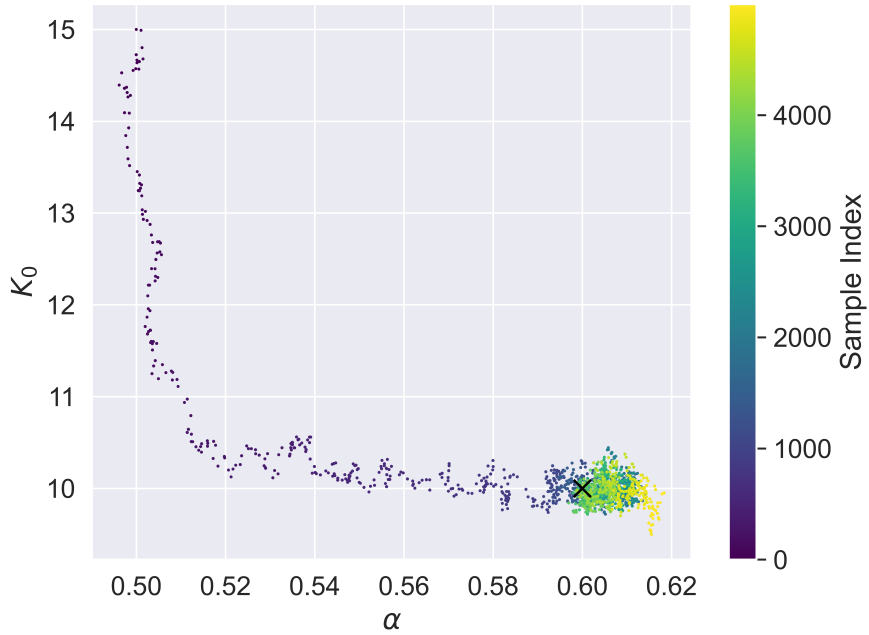


Figure 4.3: Scatter plot of the RWMH chain projected onto the (α, K_0) plane. Points are coloured by sample index: early samples (dark) show the chain traversing a low-density region far from the true parameters (black cross), while later samples (light) concentrate near the posterior mode. The extended transient illustrates the cost of burn-in when each likelihood evaluation requires a full forward solve.

A more efficient strategy would be to first locate the high-density region of the posterior — effectively maximising the likelihood — and then initialise the Markov chain there, so that all samples contribute to the posterior approximation from the outset. Gradient-based optimisation methods are well suited to this task, as they exploit the local slope of the likelihood surface to move rapidly toward the mode. More ambitiously, gradient information can be incorporated into the MCMC proposal itself, yielding samplers that explore the posterior far more efficiently than the random walk. Both of these ideas require the ability to differentiate the log-likelihood with respect to the parameters, which in turn requires differentiating through the forward map $\mathcal{P}$. We develop this capability in the following chapter.

Chapter 5

Differentiating Through the Forward Solver

The previous chapter postulated that gradient information is could be desirable both for initialising the Markov chain and for constructing more efficient MCMC proposals. Computing the gradient of the log-likelihood Eq. (4.7) with respect to the parameters ϕ requires differentiating through the forward map $\mathcal{P}$, which itself involves solving a sequence of tridiagonal linear systems. In this chapter we describe how these derivatives are obtained efficiently.

5.1 Discretise Then Optimise

There are broadly two strategies for differentiating through a numerical solver. In the *optimise then discretise* approach one derives the continuous adjoint equations analytically and then discretises them; in the *discretise then optimise* approach one first discretises the forward problem and then differentiates the discrete computation. Generally, the discretise then optimise is preferable as it computes the exact gradient of the quantity that is actually evaluated in code, avoiding the consistency errors that can arise when the adjoint PDE is discretised independently of the forward PDE [7]. We adopt this approach throughout.

Concretely, our forward map $\mathcal{P}$ is implemented as a sequence of algebraic operations — assembling tridiagonal systems and solving them via the Thomas algorithm at each timestep, then extracting the surface flux. The gradient we seek is therefore the derivative of a composition of these discrete operations with respect to the parameters ϕ .

5.2 Reverse Mode Automatic Differentiation

The log-likelihood is a scalar-valued function of the parameter vector. In this setting reverse mode automatic differentiation (AD) is the natural choice, as it computes the full gradient in a single backward pass regardless of the number of parameters.

Reverse mode AD decomposes the gradient computation into a sequence of *vector-Jacobian products* (VJPs). If the forward computation is written as a composition $f = f_L \circ f_{L-1} \circ \dots \circ f_1$, then the chain rule gives

$$\frac{\partial f}{\partial \phi} = \frac{\partial f_L}{\partial f_{L-1}} \cdot \frac{\partial f_{L-1}}{\partial f_{L-2}} \dots \frac{\partial f_1}{\partial \phi}. \quad (5.1)$$

In reverse mode, this product is evaluated from left to right: at each stage a row vector (the adjoint, or cotangent) is multiplied by the Jacobian of the corresponding primitive operation. The intermediate Jacobians are never formed explicitly; only the VJP of each primitive is required.

For the forward map $\mathcal{P}$ the dominant primitives are the tridiagonal solves — one per timestep. The question therefore reduces to: how should the VJP of a tridiagonal solve be implemented?

5.3 Differentiating Through Linear Solves

Consider a single linear system $Ax = b$, where A depends on the parameters ϕ and b depends on the solution at the previous timestep. A naïve approach is to record every intermediate value of the Thomas algorithm on the forward pass and replay them in reverse during the backward pass. This is what a generic AD framework will do if the Thomas algorithm is implemented as a sequence of elementary operations: the entire computational graph of the factorisation and back-substitution is stored on a tape.

For a single tridiagonal solve on n unknowns the memory overhead of taping is modest. However, the forward map involves m consecutive solves (one per timestep), and the tape must be retained across all of them because the backward pass traverses the full time history in reverse. The total memory cost therefore scales as $O(mn)$, which becomes prohibitive for fine spatial and temporal grids. Worse, the overhead of maintaining and traversing the tape degrades runtime performance even when memory is not the binding constraint.

5.4 Adjoint of the Tridiagonal Solve

A more efficient approach exploits the structure of the linear solve directly. Suppose the forward pass solves $Ax = b$ and the backward pass receives the adjoint (cotangent) $\bar{x} = \partial\ell/\partial x$. We require the adjoint with respect to b :

$$\bar{b} = \frac{\partial\ell}{\partial b} = \bar{x}^T \frac{\partial x}{\partial b} = \bar{x}^T A^{-1}. \quad (5.2)$$

This is equivalent to solving the *adjoint system*

$$A^T \lambda = \bar{x}, \quad (5.3)$$

and setting $\bar{b} = \lambda^T$. Similarly, the adjoint with respect to the matrix entries of A is

$$\bar{A} = -\lambda x^T, \quad (5.4)$$

which follows from differentiating the identity $Ax = b$ and substituting the solution of the adjoint system.

The key observation is that A^T is also tridiagonal (with the sub- and super-diagonals transposed), so the adjoint system Eq. (5.3) can be solved by the Thomas algorithm at the same $O(n)$ cost as the forward solve. Crucially, this approach requires only the matrix A and the forward solution x to be stored — not the full tape of intermediate values from the factorisation. The memory cost per timestep is therefore $O(n)$ rather than the $O(n)$ per operation of naïve taping, and the total cost across all m timesteps is reduced from taping the entire forward history to storing only the tridiagonal coefficients and solutions at each level.

In practice we implement this by defining a custom VJP rule for the tridiagonal solve in *JAX* [1]. On the forward pass the Thomas algorithm solves $Ax = b$ and caches A and x . On the backward pass the custom rule receives $\bar{x}$, solves $A^T \lambda = \bar{x}$ using the banded transpose, and returns $\bar{b} = \lambda^T$ and $\bar{A} = -\lambda x^T$. No taping of the Thomas algorithm’s internal operations is required.

5.5 Extension to Pentadiagonal Systems

The benefits of the custom adjoint become even more pronounced for higher-order discretisations that produce pentadiagonal (or wider banded) systems. A pentadiagonal factorisation involves substantially more intermediate values than a tridiagonal one, and taping the full forward computation through the banded solver would incur a correspondingly larger memory and runtime penalty. The adjoint approach sidesteps

this entirely: the transpose of a pentadiagonal matrix is again pentadiagonal, so the adjoint system is solved by a banded solver at the same $O(n)$ cost per timestep. The custom VJP rule generalises directly — one simply solves with the transposed band structure — making this a crucial step in enabling the use of higher-order spatial discretisations within the differentiable forward map.

Chapter 6

Hamiltonian Monte Carlo

Chapter 7

Heterogeneous Reaction

Chapter 8

Adsorption Reaction

Chapter 9

Conclusion

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